AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Final Examination
Fall Semester 2024/2025

Date: 16 /Jan./2024

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks:**50**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **not** allowed.

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Part 1:Multiple Choice Questions: [10 marks]

Question 1: Choose the most correct answer for each of the following:

1. The two separate modes of operation in the operating system are:
   a) Physical and logical modes
   b) User and kernel modes
   c) Supervisor and system modes
   d) None of the choices.
2. Which condition is not necessary for deadlock to occur?
   a) Mutual Exclusion.
   b) Resource Sharing.
   c) Hold and wait.
   d) None of the choices
3. Which of the following is a key benefit of using threads?
   a) Improved system performance
   b) Increased process creation time.
   c) Increased memory usage
   d) Simplified program design.
4. What is the correct order of operations for protecting a critical section using **Mutex** locks?
   a) signal() followed by wait()
   b) release() followed by acquire()
   c) wait() followed by signal()
   d) acquire() followed by release()
5. _____ can allow an OS to run on non-native Hardware:
   a) Shells.
   b) Spinlocks.
   c) DLLs.
   d) Emulation.
6. Which of the following is a disadvantage of Priority Scheduling?
   a) Inefficient CPU utilization.
   b) Requires complex hardware
   c) May lead to starvation.
   d) Simple to implement.
7. The program that is running all the time on the turned-on computer is:
   a) System program
   b) Firmware
   c) Kernel
   d) System Daemon
8. ..... can be used to prevent busy waiting when implementing a semaphore.
   a) Spinlocks
   b) Allowing the wait() operation to succeed.
   c) Mutex lock
   d) Waiting queues
9. If a system is in the unsafe mode, this means that we strictly have:
   a) Deadlock
   b) A possibility of a deadlock
   c) No deadlock
   d) Either a or b
10. Which of the following is not a synchronization tool?
    a) Mutex.
    b) Semaphore
    c) Process table.
    d) Monitor

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Part 2: Short and Essay questions: [15 marks]

Question 2: [3 marks]
What are the two types of **semaphores**?

Question 3: [4 marks]
Explain the difference between **deadlock** and **starvation**, by defining each.

Question 4: [4 marks]
Address binding of instructions and data to memory addresses can happen at three different stages.
Discuss briefly any two stages.

Question 5: [4 marks]
Explain the difference between **preemptive** and **non-preemptive** scheduling.

Part 3: Problem Solving questions: [25 marks]

Question 6: [6 marks]
Consider the following memory partitions of 600 KB, 380 KB, 100 KB, 300 and 400 KB (in order), and processes of 190 KB, 200 KB, 450 KB, and 220 KB (in order). How would each of the **best-fit** and **worst-fit** algorithms place the above processes in the given memory partitions?

Note: Use the below table of the memory partitions and fill it out for each algorithm.

Memory Partition	Process
600	
380	
100	
300	
400	

Question 7: [6 marks]
Calculate the Effective Access Time (**EAT**) in nanoseconds and milliseconds for a demanding page system, where Memory access time is 190 nanoseconds, the average page-fault service time is 10 milliseconds, and the page fault, **p**, is 0.2/100. (Show the full steps of your calculations)

N.B: EAT = (1 – *p*) x memory access+ *p* (average page fault services).

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Question 8: [6 marks]

Consider the following page reference string: 5,4,2,3,4,1,6,2,3,7

Ref String	5	4	2	3	4	1	6	2	3	7
f1										
f2										
f3										
H/PF										

Assume that there are **three frames**, and the **Optimal algorithm** was used.
Answer the following questions:

a- Fill in the table above after copying it to your answer booklet. (5 marks)

b- Find the number of Page Faults (PF). (1 mark)

Instructions:

- In the last row, you should indicate whether we have a hit (**H**) or a page fault (**PF**)
- In the last column, and since you do not have future pages, use **FIFO**.

Question 9: [7 marks]

Assume having four processes P1, P2, P3 and P4, and the resources R1,R2, R3 and R4 - each having only 1 instance.
Below are the relations between the processes and resources:

- P1 → R1
- P2 → R3
- R1 → P2
- R2 → P3
- R3 → P4
- P3 → R3
- R4 → P1

a) Draw the corresponding Resource-Allocation Graph (RAG). (4 marks)

b) Convert the above RAG into a **wait-for** graph. (2 marks)

c) Determine if there is a deadlock and justify your answer. (1 mark)

End of questions

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